COURSE 2 · SECTION 8 · WORKED SOLUTION

Change cycle time and backlog

Section 8 (videos S08.1 and S08.2) · Engineering change management

DESIGN QUESTION

How long does an engineering change take from request to close, how big is the open backlog at a given date, what do changes cost, and how did ECR-3200 compare with a typical change?

Steps

1. Open Section09_1_Start.pbix (or Engxtech_S08_2.pbix) > File > Save as > Engxtech_S08_Exercise.
2. Changes page: cards Median Cycle Days, P90 Cycle Days; the stage-median column chart; the Open Backlog line (X axis Dates[MonthStart], categorical).
3. Cards Rework Changes, Net Change Cost (title 'Net change cost (USD)'), Added Change Cost, Change Savings.
4. Table filtered to ChangeID = ECR-3200 with the five dates, CycleDays and ApprovalDays.
5. Right-click _Measures > New measure: Changes Slower Than ECR-3200 (below); put it on a card.

DAX calculated columns (Changes, from the video)

```
CycleDays =
IF ( NOT ISBLANK ( Changes[CloseDate] ), DATEDIFF ( Changes[RequestDate], Changes[CloseDate], DAY ) )
ApprovalDays =
IF ( NOT ISBLANK ( Changes[ApprovalDate] ), DATEDIFF ( Changes[TriageDate], Changes[ApprovalDate], DAY ) )
```

DAX measures (_Measures, from the video, plus one new)

```
Median Cycle Days =
MEDIANX ( FILTER ( Changes, Changes[Status] = "Closed" ), Changes[CycleDays] )
P90 Cycle Days =
PERCENTILEX.INC ( FILTER ( Changes, Changes[Status] = "Closed" ), Changes[CycleDays], 0.9 )
Median Approval Days = MEDIAN ( Changes[ApprovalDays] )

Open Backlog =
VAR d = MAX ( Dates[Date] )
RETURN IF ( MIN ( Dates[Date] ) > DATE ( 2026, 9, 30 ), BLANK (),
    CALCULATE ( COUNTROWS ( Changes ), REMOVEFILTERS ( Dates ),
        Changes[RequestDate] <= d && ( ISBLANK ( Changes[CloseDate] ) || Changes[CloseDate] > d ) ) )

Rework Changes    = CALCULATE ( COUNTROWS ( Changes ), Changes[ParentChangeID] <> "" )
Net Change Cost   = SUM ( Changes[CostImpact_USD] )
Added Change Cost = CALCULATE ( [Net Change Cost], Changes[CostImpact_USD] > 0 )
Change Savings    = CALCULATE ( [Net Change Cost], Changes[CostImpact_USD] < 0 )

-- new in this exercise
Changes Slower Than ECR-3200 =
CALCULATE ( COUNTROWS ( Changes ), Changes[Status] = "Closed", Changes[CycleDays] > 123 )
```

PERCENTILEX.INC matches Excel PERCENTILE.INC (linear interpolation), so p90 = 83.7 d can fall between two real cycle times. The median says a typical change closes in about 47 days; ECR-3200 took 123 days, longer than 9 in 10 closed changes, and most of the extra time was in approval (57 days against a median of 13). Rework Changes uses <> "": in DAX an empty text and a blank both compare equal to "".

Expected result

Check	Result
Median closed cycle time	**47.0 d**
P90 closed cycle time	**83.7 d**
Median approval stage	**13.0 d**
Open backlog as of 31 Dec 2025	**15**
Open backlog as of 30 Sep 2026	**24**
Rework changes	**9**
Net / added / savings (USD)	**158,004 / 251,790 / -93,786**
ECR-3200 cycle / approval stage	**123 d / 57 d**
Closed changes slower than ECR-3200	**1**